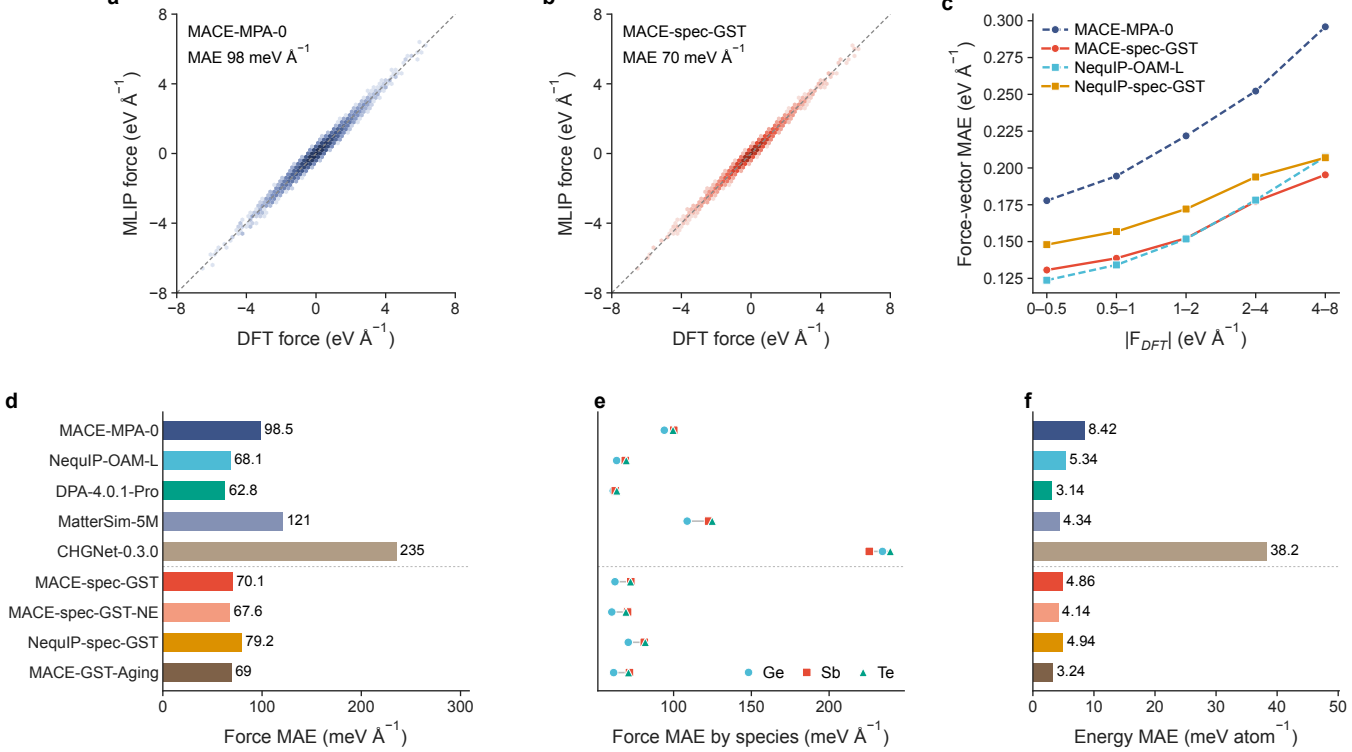
a
MACE-MPA-0
MAE 98 meV Å^{-1}
MLIP force (eV Å^{-1})
DFT force (eV Å^{-1})
b
MACE-spec-GST
MAE 70 meV Å^{-1}
MLIP force (eV Å^{-1})
DFT force (eV Å^{-1})
c
MACE-MPA-0
MACE-spec-GST
NequIP-OAM-L
NequIP-spec-GST
Force-vector MAE (eV Å^{-1})
|F$_{DFT}$| (eV Å^{-1})
d
MACE-MPA-0 98.5
NequIP-OAM-L 68.1
DPA-4.0.1-Pro 62.8
MatterSim-5M 121
CHGNet-0.3.0 235
MACE-spec-GST 70.1
MACE-spec-GST-NE 67.6
NequIP-spec-GST 79.2
MACE-GST-Aging 69
Force MAE (meV Å^{-1})
e
Ge
Sb
Te
Force MAE by species (meV Å^{-1})
f
8.42
5.34
3.14
4.34
38.2
4.86
4.14
4.94
3.24
Energy MAE (meV atom^{-1})